

Tracing Model-Generated DNA with Position-Independent SynthID Watermarking

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Abstract. Genomic language models can generate DNA sequences that resemble naturally occurring genomic material, yet the resulting sequences contain no inherent record of their origin. A watermark embedded during sequence generation can enable a holder of the corresponding secret key to identify model-generated output subsequently. In DNA, however, unknown generation boundaries, token alignment, and strand orientation complicate detection, particularly when generated sequence is embedded within a longer read.

We integrate the SynthID tournament watermark into Carbon-500M and GENERator-v2 1.2B, two genomic language models that generate non-overlapping six-base tokens, and evaluate sampler fidelity, sequence-quality proxies, and position-independent detection. Neither model exhibited a statistically detectable difference in model likelihood or the pre-specified sequence measures after multiple-testing correction. The verifier requires the DNA, the key, and fixed public settings, and controls the read-level test through a correction over both strands, every starting base, and four window lengths. In each model, it detected all 384 held-out marked reads in the unedited condition and after each tested single-base substitution, insertion, or deletion. Ordinary and wrong-key controls yielded at most one positive per model, control family, and condition. Carbon window-level analyses support recovery of token alignment through the search after an insertion or deletion. These findings support the portability of the implementation across the two evaluated models. They establish statistical detection within the tested corpus, without establishing biological function or resistance to secret-key recovery. Each model was evaluated in a single execution; independent confirmatory replication remains outstanding.

1 Introduction

Genomic language models increasingly support the generation of coding sequences, regulatory elements, and extended genomic sequences [24, 27, 3]. As these capabilities develop, establishing the provenance of generated DNA becomes an important component of responsible sequence design and synthesis. A nucleotide sequence alone does not inherently specify whether it originated from a biological sample, a computational design process, or a generative model. Once a sequence is separated from its associated metadata, records maintained by the generating software may no longer be available to downstream users. Provenance mechanisms that remain associated with a sequence throughout design and synthesis have consequently attracted attention [2].

Watermarking provides one approach to this problem by embedding a statistical signal during generation. A verifier with the corresponding secret key can subsequently test for that signal without access to the generating model, the prompt, or generation logs. Adapting language-model watermarking to DNA introduces a specific alignment problem. For a model that generates non-overlapping six-base tokens, the nucleotide sequence does not explicitly encode the token boundaries. An unknown generation boundary therefore introduces six possible token alignments. Reverse complementation adds uncertainty in strand orientation, and embedding generated sequence within a longer read introduces uncertainty in its location and extent. Detection must account for all searched alternatives when controlling the read-level

false-positive rate; a significant result at a known alignment need not remain significant after this correction.

Here, we integrate the SynthID tournament watermark [8] into Carbon-500M [3] and GENERator-v2 1.2B [22]. These models share six-base tokenisation but differ in size, training corpus, and tokenizer construction. We evaluate two questions: whether watermarking produces detectable changes in each model’s likelihood and a pre-specified panel of sequence statistics, and whether a verifier can detect the mark without knowing the generation boundary, strand orientation, or token alignment. The verifier searches both strands, every starting base, and multiple window lengths, with a single read-level multiple-testing correction covering the complete search.

Neither model exhibited a statistically detectable change in the evaluated quality proxies. The verifier detected every held-out marked read in both models, including after each tested single-base substitution, insertion, or deletion. Ordinary and wrong-key control counts were compatible with the declared false-positive target, although their confidence intervals do not establish an operational rate below that target. In Carbon, retained window-level results further show substantial separation from the corrected threshold and a shift toward shorter detection windows after insertions and deletions. These findings support application of the same sampler and verifier to two genomic models; they do not establish equivalence between the models or their generated sequences.

The analysis distinguishes mathematical identities of the watermark construction, cryptographic modelling assumptions, empirical detection results, and sequence-

based quality proxies. In particular, preservation of the model distribution is an expectation over fresh keyed functions; sampling at a fixed key and context follows a deliberately reweighted distribution. Model likelihood and sequence statistics do not establish biological function, viability, or safety, and the experiments do not evaluate resistance to secret-key recovery. Table 1 defines the terminology and notation used throughout the paper.

2 Background

2.1 Watermarking designed DNA

DNA watermarking predates its application to generative models. The first synthetic bacterial genome included watermark sequences encoding identifying information to distinguish the synthetic chromosome from its natural counterpart [12]. DNA-Crypt formalised message embedding through synonymous codon choices and error correction [16]. Related approaches embed information within an open reading frame while preserving the encoded amino acid sequence, with experimental evidence that embedded messages can persist during viral replication [23].

These approaches rely on explicit sequence design, through either insertion of a payload or modification of selected nucleotides. Their application requires suitable locations for embedding and consideration of the biological consequences of those changes. Generation-time watermarking addresses a complementary setting: the mark is introduced through the model’s sampling distribution as the sequence is generated, without requiring a separately allocated payload region. A fixed, recognisable marker also differs from a distributed keyed signal in its susceptibility to targeted removal or copying.

2.2 Watermarking language-model output

Language-model watermarking modifies token sampling to introduce a statistically detectable pattern. A widely studied approach partitions the vocabulary into pseudo-random green and red lists, increases the probability of green-list tokens, and detects the mark through their excess frequency [20]. This modification changes the sampling distribution and can introduce a tradeoff between detection strength and generation quality.

Other constructions preserve the model distribution in expectation over fresh keyed randomness. Cryptographic formulations establish computational indistinguishability from ordinary output under specified assumptions [6]. Practical approaches use inverse-transform or exponential-minimum sampling and align the observed output against a keyed random sequence during detection [21, 1]. Another family directly reweights the token distribution while recovering the original distribution in expectation [17, 29]; SynthID uses this general principle. Publicly verifiable constructions instead embed a signature through rejection sampling, subject to sufficient output entropy [10].

Robustness depends on the assumed adversary and permitted edits. Theoretical results identify limitations under sufficiently capable quality-preserving rewriting attacks [30]. Empirical studies demonstrate evasion through adaptive or recursive paraphrasing [9, 26], watermark forgery and removal through model queries [19], and recovery of keyed sequences through adaptive prompting [25]. Reuse of keyed randomness can also compromise distributional preservation [28]. Countermeasures for detecting spoofed watermarks have been investigated [13]. The present study evaluates statistical detection under single-base edits and does not test these adversarial capabilities.

2.3 SynthID

We use the binary tournament construction of SynthID-Text [8]. At each generation step, a secret key and a short context of recently generated tokens determine binary mark bits for each candidate token and tournament layer. Each layer favours candidates with a mark bit of 1. Averaged over fresh keyed functions, the tournament preserves the model’s next-token distribution; at a fixed key and context, it samples from a reweighted distribution. Detection reconstructs the mark bits from the output and tests for an excess of 1s.

Two properties motivate this construction for genomic sequences. First, the mark bits depend on local token context rather than absolute position, allowing detection to use contexts retained when a generated region is re-located or cropped. Second, the detector requires the output, key, and public settings without access to the generating model or its probabilities. SynthID-Text has also been evaluated in a large-scale text deployment [8].

2.4 Genomic tokenisation and alignment

Genomic language models support both representation learning and sequence generation [7, 24]. Several generative models, including both evaluated here, tokenise DNA into non-overlapping six-base tokens to reduce token-sequence length [27, 3, 22]. Some models additionally obtain single-base probabilities by marginalising the token distribution [22]. The released GENERator checkpoint includes a base-by-base sampling path [11]; the present study instead uses its direct canonical six-base token distribution, as specified below.

Six-base tokenisation introduces alignment uncertainty because token boundaries are not explicitly represented in the nucleotide sequence. A verifier must therefore consider all six possible alignments, both strand orientations, and the possible locations and lengths of a generated region. Selecting the strongest window without correcting for this search would overstate the statistical evidence. We consequently evaluate detection using a read-level correction over the complete search.

Recent work on genomic watermarking uses synonymous codon substitution to mark designed coding sequences and propagate the mark into translated proteins [31]. Related work applies keyed reweighting to pro-

Table 1: Terminology and notation used throughout the paper.

Word or symbol	Meaning
model	either of the two used here, each restricted to its 4,096 six-base DNA tokens
Carbon	Carbon-500M
GENERator	GENERator-v2-eukaryote-1.2b-base
token	a non-overlapping DNA sequence of six bases
prompt	a 384-base RefSeq sequence supplied to the model
continuation	the 3,072 bases generated by the model
read	what the verifier receives: prompt and continuation, $N = 3,456$ bases
mark	the SynthID tournament watermark
key, k	the secret used for watermark embedding and detection
published key	a key released with this study so the runs can be reproduced
mark bit, g	a keyed value of 0 or 1 for one candidate token in one context
layer, m	one keyed round of the tournament; $m = 30$ layers are applied
context	the four tokens generated immediately before the current one
p	the model’s probability for a candidate token
p_k	the probability the sampler actually uses, after the m layers
window	a contiguous region of the read scored by the verifier
L	window length in bases; $L \in \{384, 768, 1,536, 3,072\}$
M	windows searched in one read; $M = 16,136$ for $N = 3,456$
n	mark bits scored inside a window
S	the number of scored bits equal to 1
P_{win}	the binomial-tail probability for one window
P_{read}	the read-level P -value after correction over all searched windows
window strength	$-\log_{10} P_{\text{win}}$, so larger means stronger evidence
recall	proportion of marked reads with a positive detection
false-positive rate	proportion of control reads with a positive detection
α	the target false-positive rate for one read; $\alpha = 0.01$
ordinary	sampled directly from p , without a key
draw	one of the two continuations generated per prompt in each model

tein generation to support local verification of authorised provenance [4]. Our setting addresses detection in DNA without assuming that the generated region is coding, that a reference alignment is available, or that its boundary is known. We evaluate an established watermark construction under these conditions using only the observed read, a key, and public detector settings.

3 Methods

3.1 Model and sampling

Generation used Carbon-500M [3, 18] and GENERator-v2-eukaryote-1.2b-base [22, 11]. Both are decoder-only genomic language models that tokenise DNA into non-overlapping six-base tokens and differ in size, training corpus, and tokenizer construction. Both models were evaluated separately under the same core generation, sampler-validation, sequence-quality, and detection procedures. Results are reported per model.

At each step the model’s logits were restricted to the 4,096 canonical DNA tokens and normalised once, which for GENERator means its direct canonical token distribution rather than the base-by-base path its released code uses by default. Temperature was 1.0 and no top- k or top- p truncation was applied. The ordinary arm samples from that distribution directly and uses no key. The marked arm applies the tournament described below and then samples.

Generation ran on GPU hardware. Model scoring ran on CPU for Carbon and on GPU for GENERator, and detection ran on x86-64 CPUs for both.

3.2 Watermark embedding

At each generation step, the model assigns probability p to each of 4,096 tokens. The key, the preceding four tokens of context, and the candidate token determine its binary mark bit g . Mark bits are derived with HMAC-SHA-256 over the key, a fixed domain label, the context, and the candidate token. The tournament formula, the number of layers, the context width, and the repeated-context rule follow the public reference implementation of SynthID-Text [14]. One tournament layer reweights the distribution as

$$p'(x) = p(x) \left(1 + g(x) - \sum_y p(y) g(y) \right), \quad (1)$$

so that candidates with a mark bit of 1 gain probability according to the total probability mass assigned to such candidates. Iterating this update is equivalent to evaluating the binary tournament without explicitly constructing its 2^m leaves. The sampler applies $m = 30$ layers, each with its own mark bits, and samples a token from the resulting distribution p_k (Figure 1a).

The first four generated tokens are sampled ordinarily so that subsequent watermark contexts can be reconstructed from the continuation alone. A repeated four-token context is also sampled ordinarily and excluded

from the detector score, following the original construction. A history of 1,024 contexts is maintained for repetition detection.

Averaged over fresh keyed functions, Equation 1 leaves the model’s next-token distribution unchanged. This identity does not imply equality at a fixed key and context, where the sampler follows the explicitly reweighted distribution p_k . Sampler fidelity was therefore evaluated against p_k .

3.3 Position-independent detection

The verifier receives the read, the key, the generation domain label, and fixed public detector settings. It does not receive the prompt, the model, the model’s probabilities, the seed, the strand orientation, the token alignment, or the generation boundary (Figure 1b).

For each of the four window lengths L that fit within the read, the verifier evaluates every starting base on the supplied strand and its reverse complement. This search covers candidate generation boundaries and all six token alignments. Within each window, DNA is partitioned into tokens relative to the window start. The first four tokens provide context; the verifier reconstructs mark bits for the remaining eligible tokens and obtains S ones among n scored bits.

Under the null assumption that mark bits behave as independent fair binary values, the window-level P -value is the exact binomial tail

$$P_{\text{win}} = \Pr[\text{Binomial}(n, \frac{1}{2}) \geq S]. \quad (2)$$

The verifier scores $M = 2 \sum_L (N - L + 1)$ windows, which is 16,136 for a 3,456-base read, and reports a single read-level result:

$$P_{\text{read}} = \min(1, M \cdot \min_{\text{windows}} P_{\text{win}}), \quad (3)$$

mark found $\iff P_{\text{read}} \leq \alpha$.

The Bonferroni correction by M accounts for all searched windows, including overlapping windows, without requiring independence between their tests. The threshold uses the pre-specified $\alpha = 0.01$ and is not fitted to the evaluated reads. Equivalently, a read is positive only if at least one window reaches strength $\log_{10}(M/\alpha) = 6.21$. Window lengths are 384, 768, 1,536 and 3,072 bases; a window whose length does not fit the read is not searched, and a read shorter than every pre-specified length is not scored. Reverse-complement coordinates are mapped back onto the supplied read, and log-probabilities are retained throughout to avoid floating-point underflow.

3.4 Prompts and generation

Both models use the same cohort of 256 prompts: 384-base forward-strand windows drawn from 16 accession-versioned RefSeq chromosome records across eight model organisms. Windows were chosen by hashing a fixed label, the accession, and a segment index, before generation. Human, patient, controlled-access, organelle, plasmid, and ambiguous-base spans were excluded. No

prompt was added, moved, or removed after any model was run.

In each model, each prompt produced two marked and two ordinary continuations of 512 tokens (3,072 bases), so each model contributes 512 marked and 512 ordinary continuations. The two draws use different replay seeds in both arms, and the two marked draws use two different keys. Both keys are published for reproducibility. Deployment would require independently generated secret keys shared only with authorised generators and verifiers.

3.5 Sampler validation

For each model and prompt we fixed the model at token index 64 of its ordinary draw-zero trajectory and drew 5,000 samples per arm from that one state. Counts were compared with the exact intended law by a Monte Carlo goodness-of-fit test with 999 replicates. The marked arm was compared against p_k , not against p , because the distribution-preservation identity does not hold at a fixed key and context. Eight additional states per model used a deliberately incorrect sampler as a negative control.

3.6 Sequence measures

The model’s score is the teacher-forced negative log-likelihood per generated token under the same model and token restriction used to generate, with the prompt supplied as context and excluded from the loss. Each model scores its own outputs, so the absolute scores are not used for comparisons between models. Thirteen additional measures were specified before analysis: GC content, purine content, CpG content, base, dinucleotide and trinucleotide entropy, distinct 6-mer fraction, longest and mean single-base run, Jensen–Shannon drift from the prompt at $k = 1, 2, 3$, and perplexity. All 14 form one multiple-testing family under Benjamini–Hochberg correction.

3.7 Edits and statistics

Each edited read contains exactly one substituted, inserted, or deleted base within the continuation, at a position and with a replacement base drawn by a fixed random rule that does not consult the verifier. Prompts are the independent unit; the two draws from a prompt are paired repetitions. Uncertainty on the sequence measures comes from an equal-weight bootstrap over prompts with 20,000 replicates and a paired sign-flip test with 100,000 replicates. Detection intervals are exact two-sided binomial intervals at 95% over the 192 held-out prompts. For control families, a prompt is positive if either draw is positive; marked-output recall is also assessed by requiring both draws to be detected. All reported read-level P -values include the correction over every searched strand, starting base, and window length.

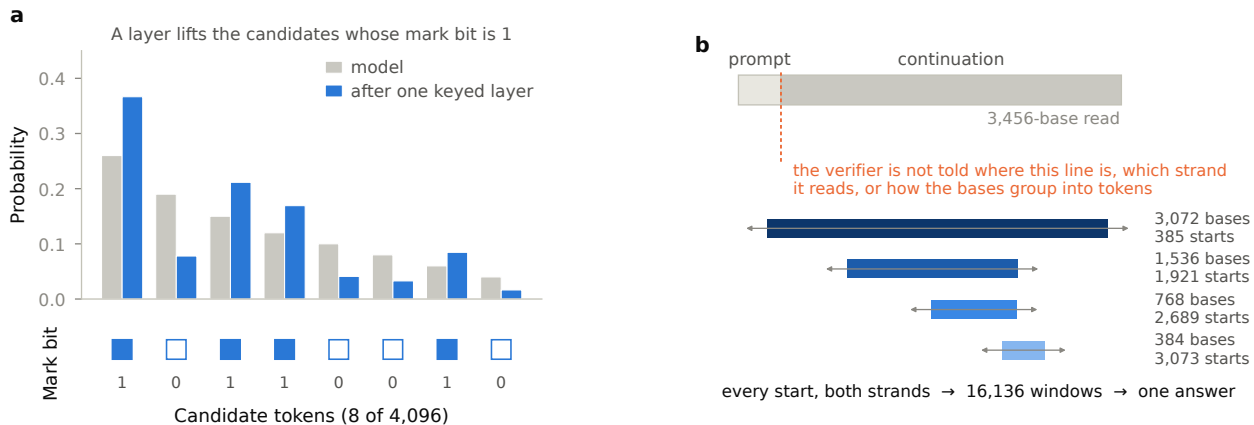


Figure 1: SynthID embedding and position-independent detection. **a**, One keyed layer, drawn for eight of the 4,096 candidate tokens with an illustrative distribution. Candidates whose mark bit is 1 gain probability; the others lose it. The sampler applies 30 such layers. **b**, Detector inputs and search. The verifier receives the read, key, and fixed public settings without the generation boundary, strand orientation, or token alignment. It scores windows of four lengths at every starting base on both strands, comprising 16,136 windows for a 3,456-base read, and reports one read-level result corrected for the full search. Panel **a** illustrates Equation 1 and is not a measurement.

4 Results

4.1 Sampler fidelity

We first evaluated agreement between each sampler and its intended distribution. Of 256 marked states, 13 fell below the nominal 0.05 level in Carbon and 5 in GENERator, against 12.8 expected by chance in each model; the ordinary arms gave 12 and 14. No arm in either model produced a rejection after correction, while all eight states evaluated with a deliberately incorrect sampler were rejected in each model. These results support sampler fidelity at the 256 evaluated states and the resolution provided by 5,000 draws per state. They do not constitute a proof of correctness at all possible model states.

4.2 Sequence-quality proxies

For every prompt and model we generated two marked and two ordinary continuations of 3,072 bases under the same conditions, giving 512 matched pairs from 256 prompts in each model.

In Carbon the model’s own score was 7.53386 nats per token for ordinary output and 7.54238 for marked output. The paired difference is +0.00852 nats per token, with a 95% interval from -0.04426 to $+0.05973$ and $P = 0.75$ (Figure 2a); as a standardized effect that is 0.0201. In GENERator the score was 7.80029 for ordinary output and 7.79465 for marked output, a paired difference of -0.00564 nats per token, with a 95% interval from -0.05614 to $+0.04378$, $P = 0.83$, and a standardized effect of -0.0139 . The paired differences have opposite signs, and neither confidence interval excludes zero.

The thirteen further measures, covering base composition, k -mer entropy, distinct 6-mers, single-base runs,

CpG content, and compositional drift from the prompt, also show no statistically detectable differences in either model. All 14 intervals contain zero, no comparison is significant after multiple-testing correction, and the smallest adjusted P -value is 0.85 in Carbon and 0.73 in GENERator. For Carbon the largest effect in absolute value is 0.084 standard deviations (Figure 2b); the equivalent summary was not retained for GENERator, for which the supported conclusion is the absence of significant differences after correction.

These results indicate no measurable quality loss under the evaluated proxies. They do not establish equivalence: the model-score confidence intervals permit changes on the order of 0.05 nats per token. Moreover, model likelihood and sequence statistics do not measure biological function, and absolute likelihood scores are not directly comparable between the two models.

4.3 Recall and false-positive rate

The prompt split was specified before examining detection results: 64 prompts were reserved for selection between two window-score summaries, and 192 were held out for evaluation. Neither subset was used to fit the threshold. For each model, the 384-base prompt was prepended to each held-out continuation, producing a 3,456-base read whose generation boundary was withheld from the verifier. Three detection families were evaluated: marked reads with the correct key, ordinary reads, and marked reads with the other draw’s key. Each family was evaluated unedited and after exactly one substitution, insertion, or deletion. The search depends on read and window lengths and is identical across models. Unedited reads require $M = 16,136$ windows and a threshold of 6.21; the window count is recomputed for the read length after an edit.

In both models, the verifier detected 384 of 384 marked

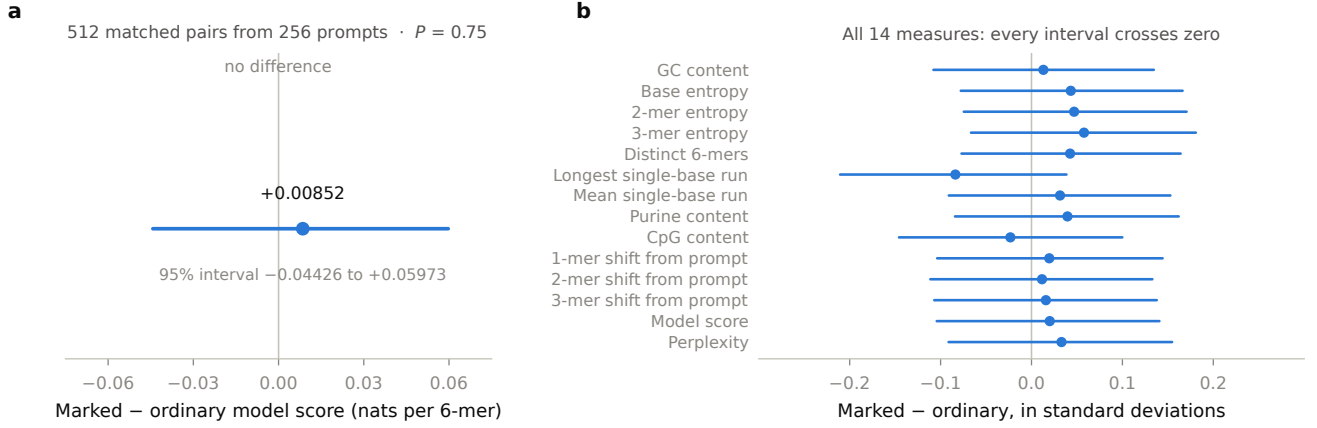


Figure 2: Model likelihood and sequence-quality proxies for Carbon. All panels are Carbon; the GENERator summaries are reported in the text only. **a**, Paired difference in the model’s score between marked and ordinary continuations, over 512 matched pairs from 256 prompts. Positive values indicate lower model likelihood for marked continuations. The bar denotes the 95% interval from a bootstrap over prompts. **b**, All 14 pre-specified measures on one scale, as the paired difference in standard deviations, with confidence intervals rescaled to the same units. Every interval includes zero, and none of the 14 comparisons is significant after multiple-testing correction.

reads in all four conditions (Table 2), a recall of 100% per model. The control positives occurred in different families. In Carbon, one of 384 ordinary reads was positive and no wrong-key read was positive. In GENERator, no ordinary read was positive and one of 384 wrong-key reads was positive in the unedited, substituted, and inserted conditions but not the deleted one. Each of those single positives is a false-positive rate of 0.26% in its cell. We report recall and the two false-positive rates separately because pooled accuracy would depend on the chosen proportions of marked, ordinary, and wrong-key reads in the evaluation cohort.

Carbon showed substantial separation between marked and control reads. On unedited reads, the weakest marked read reached window strength 124.8, while the strongest ordinary read reached 6.60 and the strongest wrong-key read reached 5.69, against a threshold of 6.21 (Figure 3a). The single ordinary read that was positive exceeded the threshold slightly. This observation is compatible with occasional false positives under the declared $\alpha = 0.01$, but does not independently establish calibration. The window-strength distribution was not retained for GENERator, so threshold margins and the edit-specific strengths below are reported for Carbon only.

Confidence intervals account for paired draws by using the prompt as the independent unit. In each model, all 192 prompts had both marked draws detected, giving an exact 95% interval of 98.10–100%. In Carbon one prompt of 192 produced an ordinary positive, giving 0.013–2.868%, and none produced a wrong-key positive, giving an upper bound of 1.903% (Figure 3b). In GENERator, no prompt produced an ordinary positive, giving an upper bound of 1.903%, and one produced a wrong-key positive, 0.013–2.868%. Within each model, control positives across conditions refer to the same prompt and draw. Carbon’s ordinary read remained positive in all four con-

ditions because each edit occurred outside its strongest window. GENERator’s wrong-key read remained positive after substitution and insertion, but not after deletion.

GENERator also provides a diagnostic assessment of the analytic null under known token alignment. Scoring ordinary output over the full 256-prompt cohort, the number of prompts producing a positive was 2, 6, 6 and 4 at the four window lengths, against about five expected at each, with exact two-sided P -values of 0.25, 0.87, 0.76 and 0.84. The analytic null is therefore compatible with the ordinary-output counts at each evaluated length. This diagnostic was performed for GENERator only and does not replace the full-search correction.

4.4 Detection after single-base edits

Window-level analyses are available for Carbon. A substitution perturbs token content and nearby watermark contexts while preserving downstream token alignment. The median window strength decreased from 790.6 for unedited reads to 774.7 after substitution (Figure 4a). Insertions and deletions shift downstream token alignment, reducing the signal in windows that span the edit.

For unedited reads, the full 3,072-base window provided the strongest evidence in 383 of 384 reads. After insertion, that count decreased to 203, with 181 reads favouring a shorter window; after deletion, the corresponding counts were 232 and 152 (Figure 4b). Searching every starting base allows windows downstream of the edit to recover the original token alignment. The observed shift toward shorter windows is consistent with detection using regions whose alignment is preserved or recovered. Median strength decreased to 431.5 after insertion and 443.7 after deletion. The weakest marked read across all conditions still reached 82.2, above the threshold of 6.21, and every marked read remained detectable.

GENERator’s retained results establish detection out-

Table 2: Positive detections for each model across 192 held-out prompts with two draws each. Each edit affects exactly one base. The final two rows of each block summarise unedited reads at the prompt level, accounting for paired draws; intervals are exact and two-sided. All marked prompts also had both draws detected in every edit condition. Results are reported separately for each model.

Edit applied to the read	Marked, right key	Ordinary	Marked, wrong key
<i>Carbon</i>			
None	384/384	1/384	0/384
One substitution	384/384	1/384	0/384
One insertion	384/384	1/384	0/384
One deletion	384/384	1/384	0/384
Prompts with a positive draw	192/192	1/192	0/192
Exact 95% interval	98.10–100%	0.013–2.868%	0–1.903%
<i>GENERator</i>			
None	384/384	0/384	1/384
One substitution	384/384	0/384	1/384
One insertion	384/384	0/384	1/384
One deletion	384/384	0/384	0/384
Prompts with a positive draw	192/192	0/192	1/192
Exact 95% interval	98.10–100%	0–1.903%	0.013–2.868%

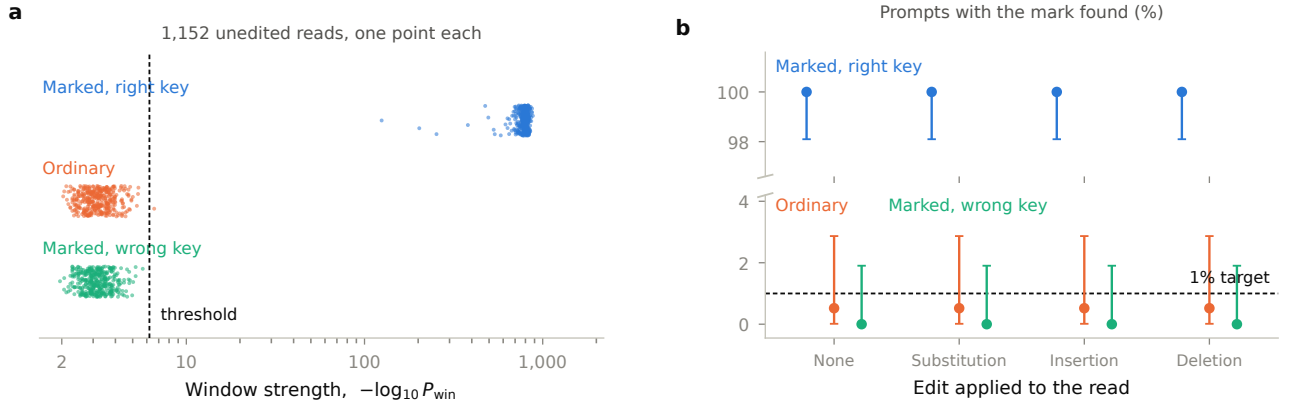


Figure 3: Position-independent detection in Carbon. All panels are Carbon; GENERator’s decisions are in Table 2, and its window strengths were not retained. **a**, Strength of the best window in each unedited read, one point per read, on a log scale. The dashed line denotes the threshold corrected for the search over 16,136 windows. All marked reads exceed the threshold; one ordinary read is also positive, whereas all wrong-key reads remain below it. **b**, Proportion of the 192 held-out prompts with positive detections, with exact 95% intervals, split across a break in the axis. Results are the same in all four edit conditions. The control intervals reach above the 1% target, so these data are consistent with a false-positive rate below 1% without establishing one.

comes without the corresponding window-level analysis. All 384 marked reads were detected in each condition, and its single wrong-key control positive was absent after deletion. Thus, both models support detection after the tested single-base edits, while the evidence concerning changes in window strength and length is specific to Carbon.

5 Discussion

This study integrated the SynthID tournament watermark into Carbon-500M and GENERator-v2 1.2B and evaluated sampler fidelity, sequence-quality proxies, and position-independent detection. Both models showed

agreement with the intended sampling distribution at the evaluated states, no statistically detectable differences in the declared quality proxies, and detection of all held-out marked reads under unedited and tested single-base edit conditions. These findings support the feasibility of generation-time watermarking and subsequent detection without access to the model or generation boundary within the evaluated corpus.

A central methodological result is that detection remained effective after explicit correction for alignment and location uncertainty. The verifier searches both strands, every starting base, and four window lengths, and includes every tested window in the read-level correction. Carbon’s window-level results show that the signal accumulated over the evaluated continuations sub-

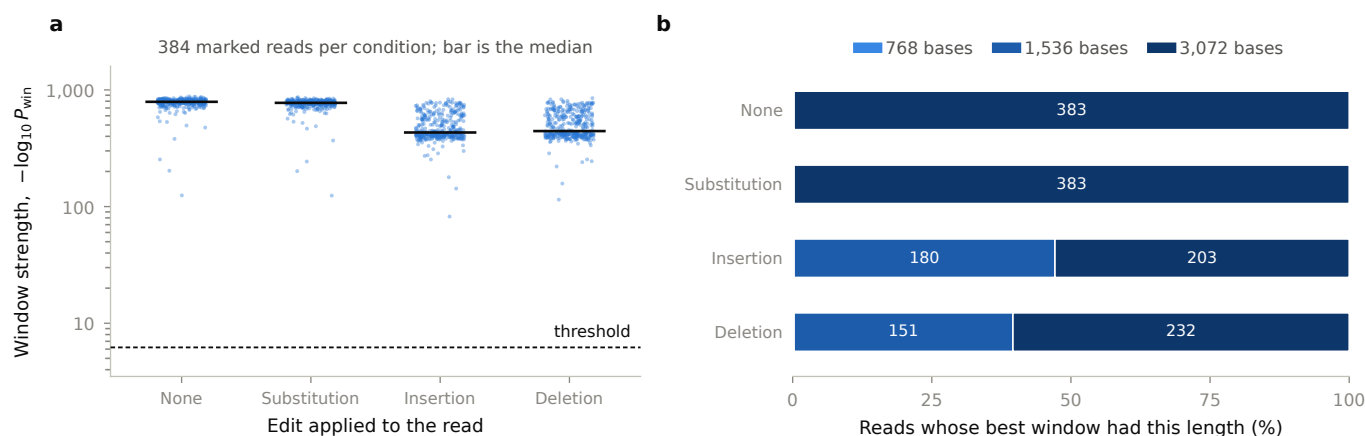


Figure 4: Effects of single-base edits on detection in Carbon. All panels are Carbon. **a**, Strength of the best window in each of the 384 marked reads per condition, on a log scale, with the median shown as a bar. The dashed line is the detection threshold. Substitution produces a small reduction in median strength; insertion and deletion reduce it by approximately half, while all marked reads remain above the threshold. **b**, Length of the window that carried the strongest evidence. Numbers inside the bars are read counts out of 384. The full 3,072-base window is strongest for nearly all unedited and substituted reads. Shorter windows are more frequently selected after insertion or deletion.

stantially exceeded the corrected threshold. This margin should not be assumed for shorter continuations or longer surrounding reads: shorter generated regions provide fewer mark bits, while longer reads increase the number of candidate windows. The minimum detectable continuation length under these conditions remains to be established.

Applying the same sampler and verifier to two models supports portability across the evaluated model implementations. The detector depends on the observed read and public settings, so its statistical procedure is unchanged between Carbon and GENERator. Their agreement in detection outcomes and the absence of measurable quality loss provide evidence beyond a single model application. However, the models share the same prompt cohort and token length, limiting extrapolation to other corpora and tokenisation schemes. This comparison also does not establish relative biological quality or equivalence between the models.

The Carbon results clarify the role of alignment search after insertions and deletions. A substitution preserves downstream token boundaries, whereas an insertion or deletion changes the alignment of every subsequent token within a window spanning the edit. The increased frequency of shorter strongest windows after these edits is consistent with the verifier selecting regions whose alignment remains usable. Exhaustive search over starting bases also permits recovery of the original alignment downstream of an edit. This mechanism supports the observed single-edit results, but provides no general guarantee against more extensive modification. Theoretical work on watermark removal further emphasises that robustness depends on the permitted adversarial capabilities [30]. Here, each read received exactly one randomly specified edit, without access to the verifier’s score.

The alignment problem is related to synchronisation errors in coding theory. Synchronisation strings explicitly

encode indexing information to support recovery from insertions and deletions [15]. Pseudorandom codes combine computational indistinguishability with decodability under specified error models, motivating their use in watermarking [5]. Our verifier addresses alignment uncertainty through exhaustive search, without adding an explicit indexing code to the sequence. The resulting cost is the number of windows evaluated and the associated multiple-testing correction. This analogy concerns the alignment problem; it does not imply the error-correction guarantees of those coding constructions.

Watermarking designed coding sequences through synonymous substitution, including approaches that preserve the mark in translated proteins [31], addresses a complementary objective. Such methods assume a coding region and permit explicit sequence modification, whereas our evaluation concerns detection in generated DNA without those assumptions. Combining these approaches may support broader provenance workflows, but their compatibility and biological consequences would require separate evaluation.

6 Limitations

The study evaluates two models, one watermark construction, one verifier, and one shared prompt cohort. Each model contributes a single execution, and independent confirmatory replication remains outstanding. Both models use six-base tokens, as do other genomic models [27]; models with single-base or variable-length tokenisation would present different alignment requirements. Longer reads would also increase the search correction. Because both models use the same prompts, the study evaluates portability across models without independent replication across corpora.

The retained analyses differ in coverage. Carbon provides window-strength distributions, standardized effects

for all 14 sequence measures, and the length of the strongest window after each edit. GENERator provides the aligned-detector null diagnostic. Claims based on these analyses are attributed to the corresponding model; the empirical panels in the four figures use Carbon results.

Generation was performed on GPU hardware and detection on x86-64 CPUs. These results have not been independently replicated across computing platforms. For the Carbon detection run, the retained protocol document differs from the protocol hash recorded in the result. This documented provenance discrepancy remains unresolved and limits verification of the executed protocol. Confirmatory replication is required to address these execution and provenance limitations.

The observed false-positive counts are compatible with the 1% target but do not establish an operational rate below it. Among 192 independent prompts, Carbon had one ordinary positive and no wrong-key positives; GENERator had one wrong-key positive and no ordinary positives in the unedited condition. The confidence interval for one positive prompt extends to 2.87%, and the upper bound for zero positives is 1.90%. More independent prompts are needed to estimate rates near the declared target precisely.

The experiments use published keys for reproducibility and therefore do not evaluate secrecy or resistance to key recovery in deployment. The analytic null also relies on the modelling assumption that mark bits behave as independent fair binary values without the correct key. Adaptive prompting has recovered keyed sequences in text watermarking [25], and model queries have enabled watermark forgery and removal [19]. Whether these attacks transfer to the evaluated genomic setting is untested. The assumed attacker does not hold the key and cannot observe or query the verifier’s score.

Finally, model likelihood and sequence statistics are biological proxies. They do not measure expression, replication, function, viability, or safety. The absence of detectable changes in these measures therefore does not establish preservation of biological function. Addressing that limitation requires experimental assays.

7 Conclusion

The SynthID tournament watermark was applied to Carbon-500M and GENERator-v2 1.2B without a statistically detectable loss in the evaluated model-likelihood and sequence-quality proxies. Position-independent verification detected all 384 held-out marked reads per model in the unedited condition and after each tested single-base substitution, insertion, or deletion, with read-level correction over both strands, every starting base, and four window lengths. Control counts were compatible with the declared false-positive target, although 192 independent prompts provide limited precision near a 1% rate. Carbon’s window-level analyses further support alignment recovery through the search after insertions and deletions. These findings support the feasibility of

statistical provenance detection in model-generated DNA within the evaluated scope, with independent replication, broader sequence coverage, and biological validation remaining necessary next steps.

Data and code availability

The two experimental keys are published as public fixtures for reproducibility. They are not deployment secrets, and the experiments do not assess resistance to secret-key recovery. The generated sequences, verifier outputs underlying the figures, prompt set, and analysis and figure code are available from the authors on request.

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